

PRELAB 2: AC INDUCTION AND INDUCTANCE

1. Introduction

This prelab guides you through building and analysing a 2D electromagnetic model of a transformer core using ANSYS Maxwell 2D (Transient solver). The goal is to help you visualise how AC induction works, how magnetic flux flows through a core, and how coil currents and voltages behave under time-varying excitation.

You will complete a simulation, generate field plots, and answer conceptual questions that prepare you for Experiment 2. Completing this prelab is worth 10 marks and will be checked by demonstrators during the lab session.

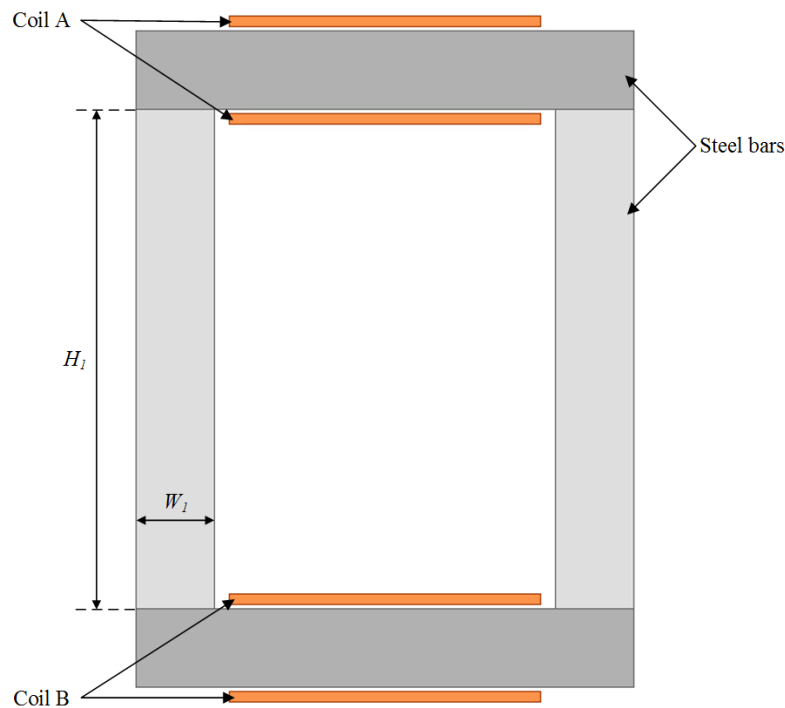


Fig.1: 2D model of the transformer with rectangular core.

2. Objectives

In this Prelab, you will:

- build a 2D transformer model with a rectangular core.

- assign materials, coils, excitations, and boundaries.
- run a transient electromagnetic simulation.
- plot and interpret magnetic fields, induced voltages, and currents.
- compare simulation outcomes with experimental results.
- write a 2-page report by answering a set of questions (see below)

3. Pre-lab Tasks

Task 1: Model setup and field visualisation

You will follow the video guide and/or simplified instructions in the Appendix to build a transformer model in ANSYS Maxwell 2D (Electromagnetic transient solver).

After solving the model,

- Generate plots of coil currents, induced voltage (Coil B), flux, and magnetic field distribution.
- Capture a clear, well-labelled screenshot of these plots for your report.
- Your report should be concise and focused on physical interpretation, not long descriptions of button clicks.

Task 2: Analysis and Discussion

These questions relate to the transformer induction and inductance simulation performed in ANSYS Maxwell. Answer the questions concisely in your **2-page pre-lab report**, focusing on **correct results, clear plots, and physical interpretation**. Be prepared to explain your answers orally to the demonstrator.

Question 1: Transient vs steady-state current

The simulation shows that the **initial current amplitude** in Coil A is higher than the steady-state current.

- Explain why this transient inrush current occurs at the start of excitation.
- Relate your explanation to magnetic flux establishment, inductance, and core behaviour.

Question 2: FEM vs experimental current

Compare the **FEM-calculated RMS current** in Coil A with the **experimentally measured RMS current**.

- State both values clearly.
- Comment on the difference between them.
- Explain possible reasons for this difference, considering modelling assumptions and experimental limitations.

Question 3: Time-domain waveforms and steady-state values

Generate and include screenshots of the following three plots:

1. **Coil A current vs time**
 - Show the instantaneous (time-domain) current waveform.
 - Overlay the corresponding **RMS steady-state current**.
2. **Coil B voltage (E2) vs time**
 - Show the instantaneous (time-domain) voltage waveform.
 - Overlay the corresponding **RMS or peak steady-state value**.
3. **Magnetic flux vs time**
 - Show the instantaneous magnetic flux waveform in the core.
 - Overlay the corresponding **maximum or steady-state value**.

Each plot must clearly display **both curves** (instantaneous value and RMS/Max steady-state value).

Using these plots:

- Comment on the variation of **Coil B voltage (E2)** and **magnetic flux**.
- Compare the simulated trends with the **experimental observations**.

Question 4: Magnetic field distribution in the core

Include a screenshot of the **magnetic field (flux density) plot**.

- Is the magnetic field evenly distributed inside the core?
- Identify regions where the field is **non-uniform**.
- Indicate where the **maximum flux density** occurs and explain why this happens.

Question 5: Field animation and core saturation

Animations can be used to visualise how the magnetic field varies with **time, current, and load**.

- Use the field animation to observe changes in the colour map.
- Explain what the colour variation represents physically.
- Identify when the core is more **saturated** and explain how saturation affects the **inductance**.

(You may close the animation once you have completed your observations.)

Task 3: Verification (Compulsory)

You **must** complete this task to receive full pre-lab marks.

Using the Maxwell design “**RectangularCore_NoLoad**”:

- Set the **air gap length** equal to the last three digits of your student ID expressed in metres.
 - Example: for student ID z5XXX577, set the gap to **0.577 m**.

Complete the following:

1. Record the calculated **inductance values** in your simulation report.
2. Take a **screenshot of the magnetic field plot at $t = 0.6$ s** and include it in your report.

5. Marking rubric (10 marks)

- **Correct simulation setup and results (6 marks)**
 - Correct model geometry, boundary conditions, and solution (3)
 - Correct field plots and extracted quantities (capacitance/charge) (2)
 - Clear presentation in a concise 2-page report (1)
- **Oral explanation and interpretation (4 marks)**
 - Clear explanation of modelling steps (2)

Correct physical interpretation of results (field behaviour, edge effects, discrepancies) (2)

Appendix: Finite Element Modelling of a Transformer

You must use **myAccess**>ANSYS Maxwell for this prelab, as the student version does not support the transient solver.

Follow the steps below to set up and solve the **2D transient transformer model** in ANSYS Maxwell. These instructions are intentionally concise; refer to the video instructions on Moodle if needed.

1. Start a New Maxwell 2D Project

- Open **ANSYS Maxwell** (same procedure as Prelab 1).
 - Create a new project: **File > New**.
 - Insert a design: **Project > Insert Maxwell 2D Design**.
 - Rename the design to **RectangularCore**.
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2. Tool and Display Settings (once only)

Go to **Tools > Options** and set:

- **General > Desktop Configuration**: enable *Electronics Pro / Premium / Enterprise licensing*.
 - **Maxwell 2D > Boundary Assignment**: enable *Use Wizards* and *Duplicate boundaries with geometry*.
 - **3D Modeler > Drawing**: enable *Edit properties of new primitives* and *Automatically cover closed polylines*.
 - **3D Modeler > Display > Rendering**: set default transparency to **0.8**. Click **OK**.
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3. Solution Type and Units

- **Maxwell 2D > Solution Type**:
 - Geometry: Cartesian (XY)
 - Solver: **Magnetic** → **Transient**
 - **Modeler > Units**: set to **mm**.
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4. Save the Project (Very important! if you do not want to lose data due to unintended interruptions)

- Save locally as **ELEC3115_Lab2_zID** (replace zID with your UNSW zID).
 - Grant Citrix access if prompted.
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5. Create the 2D Core Geometry

Create four rectangular steel bars forming a rectangular core:

- Left and right vertical bars
 1. Draw > Rectangle > select origin (0,0,0), dx:21, dy:133, dz:0
 2. Change Xsize to W1*1mm and Ysize to H1*1mm
 3. Edit > Select All
 4. Edit > Duplicate > Along Line > coordinate (0,0,0), dx:112, dy:0, dz:0
 5. Geometric properties window > Change Vector: ((H1-W1) *1mm, 0 ,0)
- Top and bottom horizontal bars
 - Repeat step 1, 2,3,4 above but use dx:133, dy: -21, dz:0 in step 2 and dx:0, dy:154, dz:0 in step 3
 - After duplication change vectors: (0, (H1+W1) *1mm, 0)

Use **parametric variables** (no units):

- W1 = 21
- H1 = 133

Assign material **steel_1008** to all core bars and rename them:

- Bar_left, Bar_right, Bar_top, Bar_btm
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6. Create Coil Geometry

- Draw two rectangular coils on the lower limb → **CoilB_1** (coordinate: (25, 1, 0) dX:83, dY:3, dZ:0); **CoilB_2**(Duplicate → (0,0,0); (dX:0, dY:26, dZ:0))
 - Duplicate them to the upper limb → **CoilA_1**, **CoilA_2**;(0,0,0) (dX:0, dY:154, dZ:0)
 - Set coil colour to orange for visibility, assign material: copper
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7. Create Simulation Region and Air Gap

- Draw a large surrounding rectangle: (-30, -61, 0) dX:193, dY:255, dZ:0)and rename it **Region**.
 - Set Region display to **Wireframe**.
 - Introduce an air gap by moving the bottom bar and Coil B using variable:
 - Select **Bar_btm**, **CoilB_1**, and **CoilB_2**
 - Edit > Arrange > Move: (0,0,0); dX: 0.0, dY: 0.0, dZ: 0.0
 - Modeler Properties window: Change Move Vector: **0, Gap*-1 mm, 0**
 - $\text{Gap} = 0.0$ (no units) (you will be able to change value later)
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8. Duplicate Design for No-Load Case

- Copy **RectangularCore** and paste it.
 - Rename the copied design to **RectangularCore_NoLoad**.
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9. Model Depth and Boundary Conditions

- **Maxwell 2D > Model > Set Model Depth: 20 mm.**
 - **Edit > Selection Mode > Edges**
 - Select all edges of Region
 - **Maxwell 2D > Boundary > Assign > Vector Potential (Value: 0)**
 - Change to the selection mode back to Objects
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10. Define Coils and Windings

- Select **CoilA_1** and **CoilA_2**
- **Maxwell 2D > Excitations > Assign > Coil**
- Assign coils: Maxwell 2D > Excitation > Coil
 - Number of Conductors: N_a (Value:600)
 - Polarity: Positive
- Repeat for **CoilB_1** and **CoilB_2**
 - Number of Conductors: N_b (Value:2400)
 - Polarity: Positive
- Select **CoilA_2** and **CoilB_2** from **Project Manager**
Change Polarity type to **Negative**.

- Create **Primary winding**
 - Maxwell 2D > Excitation > Add Winding
 - Name: **Primary_winding**
 - Type: **Voltage**
 - Check: **Stranded**
 - Initial Current: 0 A
 - Resistance: **4.5 ohm**
 - Inductance: **0 H**
 - **Voltage: $V_{rms} * \sqrt{2} * \sin(\Omega_{rad} * Time) * 1V$**

$V_{rms}: 50 \text{ V}, \Omega_{rad}: 2 * \pi * freqV, freqV: 50$
 - Add coils to Winding:
 - In Project Manager, Right click on **Primary_winding**
 - select **Add Coils**
 - Press **Ctrl** and select the terminals **CoilA_1** and **CoilA_2**
- Create **Secondary winding** by repeating the same process as above.
 - Type: current
 - Add CoilB_1 and 2.

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11. Mesh Settings

- Select **Bar_left, Bar_right, Bar_btm, and Bar_top**
 - **Maxwell 2D > Mesh > Assign Mesh Operation > On Selection > Length Based.**
 - Element Length Based Refinement window:
 - Name: **Bars**
 - Set **maximum element length: 8mm**
 - Repeat the above operations for **Region:**
 - Name: **Region**
 - Set **maximum element length: 15 mm**
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12. Analysis Setup and Solve

Maxwell 2D > Analysis Setup > Add Solution Setup

General Tab

- Stop time: **0.6 s**
- Time step: **1 ms**

Save Fields Tab

- Save fields from **0.56 s to 0.6 s** (every step)

Solver Tab

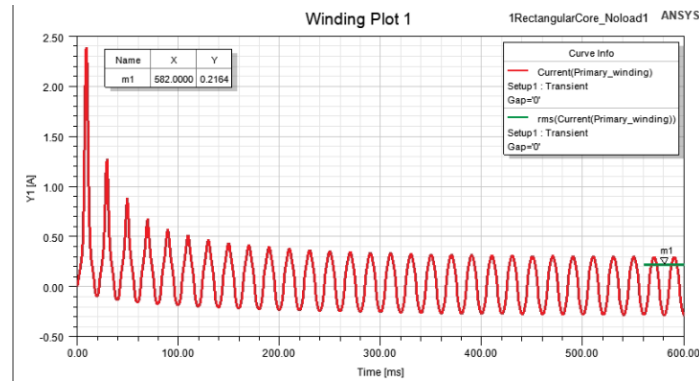
- Nonlinear residual: **0.0001**

Run **Validate**, **Analyze All**, and save the project.

13. Create Required Plots

Create a transient report for Coil A current and its RMS value

- **Maxwell 2D > Results > Create Transient Report > Rectangular Plot**
- In the **Report: Project ... New Trace(s)** window:
 - Primary Sweep: **Time**
 - X: **Check Default**
 - Category: **Winding**
 - Quantity: **Current (Primary_winding)**
 - Click **New Report** (This creates an XY plot of coil A current vs. time)
 - Don't close the window.
- In the same **Report: Project ... New Trace(s)** window:
 - Modify Y by clicking **Range Function...**
 - Category: **Math**
 - Function: **rms**
 - Over sweep: **Time**
 - Specify range: **0.56 s to 0.6 s**
 - Click **OK**
 - Modify **X**:
 - Uncheck **Default**
 - Type into the box: **Time**
 - Click **Add Trace**
 - The created plot is shown in the figure below:
 - The red curve is the real-time coil A current variation with time
 - The green line is the rms current at the steady state (The colour may be different.)



Transient report of the coil A current

- Right-click on the plot, select **Marker > Add Marker**
- Click on the green line
- Press **Esc** on the keyboard
- We can read from the plot that the steady state coil A current is 0.2164 A

Create a transient report for the Coil B voltage (E2) and flux

Repeat the above processes for:

- Coil B voltage: **InducedVoltage(Secondary_winding)**
- Peak flux: **FluxLinkage(Primary_winding)/N_a**
 - Note that the flux linkage must be divided by the turn number (**N_a**) to get the flux value.
 - To get the peak flux, you need to use the math function **max()**.
- Don't forget to set the steady state time range for RMS and max value calculations (0.56 s to 0.6 s).
- You may have to select show all objects in active view

14. Magnetic Field Plots and Animation

Go to the Geometry Model

- **Edit > Select All**
- Hold **Ctrl** and click on the **Region** to unselect it.
- **Maxwell 2D > Fields > B > Mag_B**
- In the Create Field Plot window:
 - Quantity: **Mag_B**
 - In volume: **AllObjects**
 - Click **Done**.
- Repeat the above operations to create a field plot for flux lines
 - Quantity: **Flux_Lines**
 - In volume: **AllObjects**
 - Click **Done**.
- **View > Set Solution Context**

- Solution name: **Setup1 : Transient**
- Time: **560.0 ms**
- Check: **Lower left**
- Check: **Display speed and position values**
- **Click OK.**

Animate the field plots

- **Edit > Animate...**
- In the Select Drawing window:
 - Select Mag_B1
 - Click OK
- In the **Create Animation Setup** window:
 - Name: Mag_B_AM
 - Single variable
 - Time
 - Select all time values (default)
 - Click OK.

You may close animations once observations are complete.
